

Tracking Audit Report — Executive Summary

Executive Summary

Overall tracking quality is **medium** — RMSE 0.46 m with 43% prediction-only rows; the filter coasts

noticeably but stays near truth. The biggest error source is **GPS measurement noise / bias**. Most suspicious

behavior: recurring 'sensor_dropout' anomalies (3 occurrences). Best-performing sensor: Camera; weakest: GPS.

Weakest time window: $t = 3.0\text{--}4.0$ s (mean err 0.55 m). The system relied heavily on prediction-only Kalman rows

(43% of all rows). GPS measurements appear unreliable — see the sensor-quality section.

Scorecard

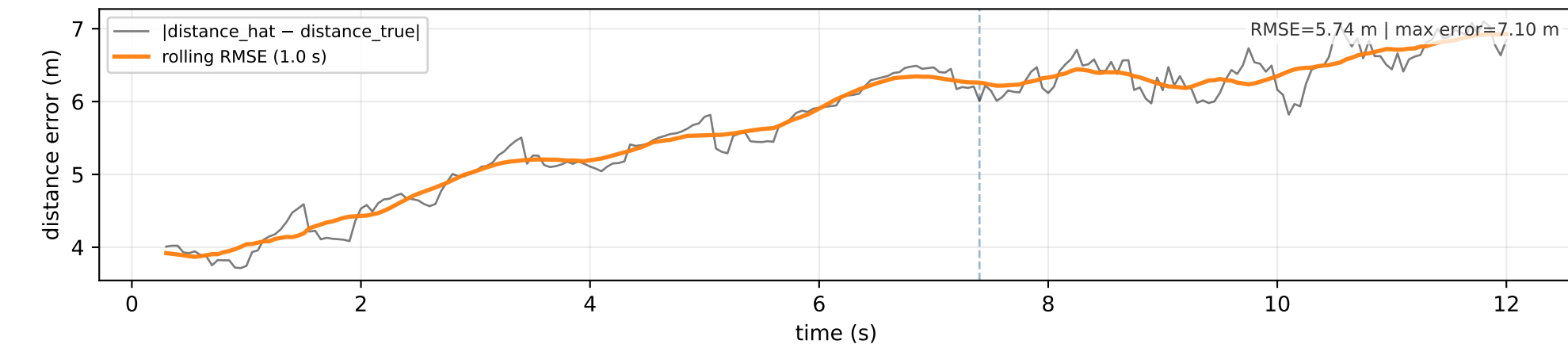
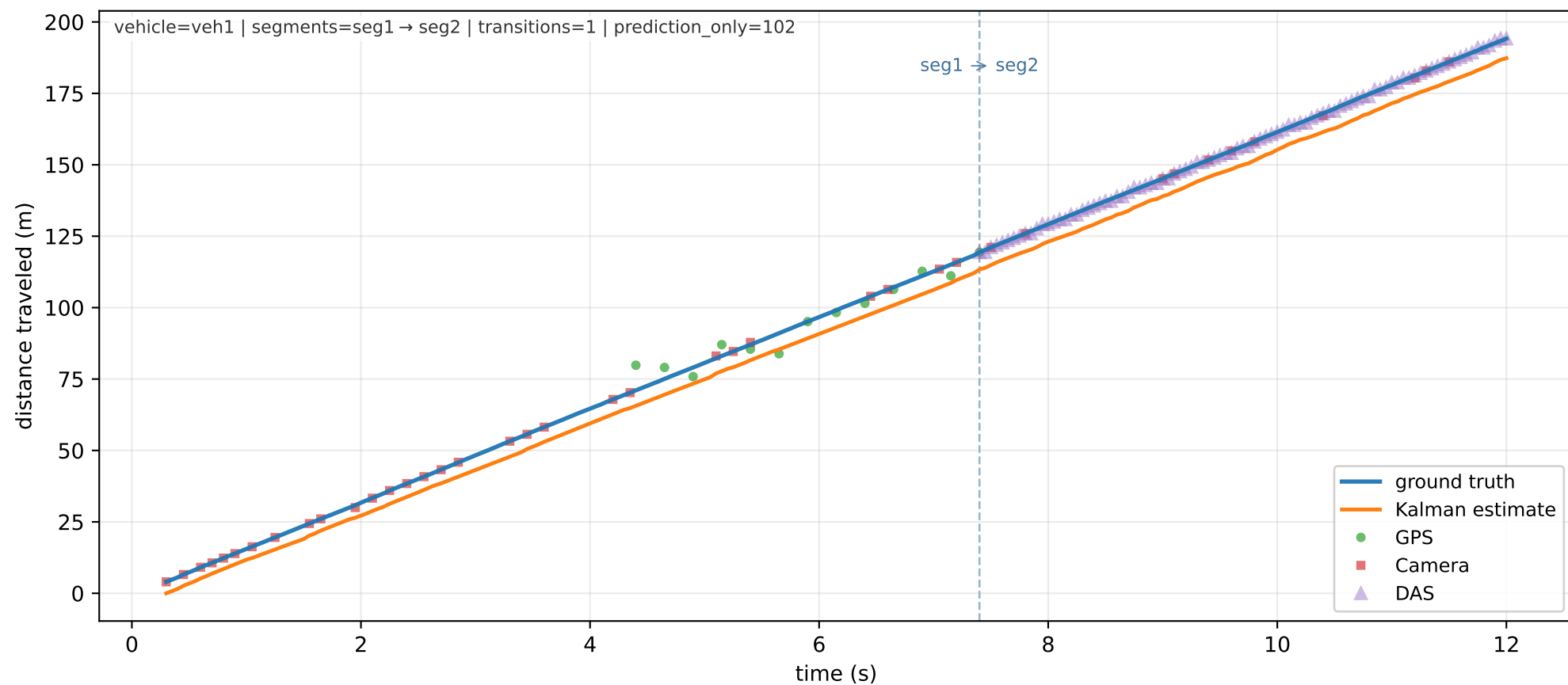
- Overall quality: **medium** — RMSE 0.46 m with 43% prediction-only rows; the filter coasts noticeably but stays near truth
- Biggest error source: GPS measurement noise / bias
- Most suspicious behavior: recurring 'sensor_dropout' anomalies (3 occurrences)
- Best sensor: Camera
- Weakest sensor: GPS
- Weakest time window: $t = 3.0\text{--}4.0$ s (mean err 0.55 m)
- Relied on prediction-only rows: yes (43% of all rows)
- Too few sensor measurements: no
- Suspicious noisy sensor: GPS
- Vehicle fragmentation suspected: no

Note: claims below are tagged confirmed / likely / hypothesis / next-check so the reader can tell facts from speculation.

Simulation Tracking Overview

Field	Value
Simulation	/sessions/nifty-dreamy-turing/mnt/final proj/outputs/_sample_audit_report
Meaningful tracks	1
Skipped tracks	0
Segment transitions	1
Total anomalies (capped)	3
Global RMSE	0.456 m
Global max error	1.620 m
Prediction-only rows	102
GPS measurements	n/a
Camera measurements	n/a
DAS measurements	n/a

Tracking trajectory — global_track_id = T000001



Segment Transitions

Track	Time (s)	From segment	To segment
T000001	7.40	seg1	seg2

Anomaly Detail

Track T000001 — sensor_dropout @ t=3.65s

Camera dropout of 1.00 s (t=3.65–6.45 s).
Before: sensors=Camera, segment=seg1, mean error=0.3 m.
During: sensors=Camera, GPS, mean error=0.5 m, 43/57 rows prediction-only.
After: sensors=Camera, DAS, GPS, segment=seg1 → seg2, mean error=0.5 m.
→ Error recovered to 0.5 m after the anomaly.

Track T000001 — sensor_dropout @ t=7.85s

Camera dropout of 1.15 s (t=7.85–9.00 s).
Before: sensors=Camera, DAS, GPS, segment=seg1 → seg2, mean error=0.4 m.
During: sensors=Camera, DAS, mean error=0.3 m.
After: sensors=Camera, DAS, segment=seg2, mean error=0.3 m.

Track T000001 — sensor_dropout @ t=9.85s

Camera dropout of 0.75 s (t=9.85–11.20 s).
Before: sensors=Camera, DAS, segment=seg2, mean error=0.3 m.
During: sensors=Camera, DAS, mean error=0.3 m.
After: sensors=Camera, DAS, segment=seg2, mean error=0.3 m.
→ Error recovered to 0.3 m after the anomaly.

Sensor Coverage Analysis

Sensor Coverage Analysis

Across 1 track(s) the audit recorded 13 GPS, 40 camera and 93 DAS measurements plus 102 prediction-only rows (43.4% of all rows). 5 time window(s) had only one sensor type active (see fragmentation / FOV sections for the likely cause).

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Sensor Coverage Analysis — table

Track	GPS	Cam	DAS	Pred	Rows	%pred	Longest meas-gap
T000001	13	40	93	102	235	43.4	0.60

Sensor Coverage Analysis — Sensor coverage looks sufficient

[INFO] Sensor coverage looks sufficient

What happened

GPS=13, Cam=40, DAS=93, prediction-only=43.4%, longest gap=0.60 s.

What it probably means

Track T000001 received a balanced mix of measurements with no large gap. The Kalman has enough evidence to stay close to truth.

Why it matters

Healthy coverage is the baseline for trustworthy tracking.

Confirmed evidence

- GPS = 13 measurements, Camera = 40, DAS = 93.
- Prediction-only rows = 43.4% of the track.
- Longest gap between real measurements = 0.60 s (track duration = 11.70 s).

Sensor Quality / Noise Analysis

Sensor Quality / Noise Analysis

Per-sensor noise / bias is computed by comparing each accepted measurement against the oracle ground truth at the same timestep. 'σ vs err' flags whether the reported sigma matches reality — a ratio far from 1.0 means the sensor model is mis-calibrated.

Sensor Quality / Noise Analysis — table

Sensor	N	Mean err (m)	Max err (m)	RMSE (m)	Bias x (m)	Bias y (m)	Mean sigma	Sigma vs err
GPS	13	3.94	9.28	4.60	0.12	0.48	2.80	$\sigma \approx \text{actual}$
Camera	40	0.39	0.88	0.44	0.04	0.06	0.32	$\sigma \approx \text{actual}$
DAS	93	0.45	1.09	0.50	-0.02	-0.04	0.30	$\sigma \approx \text{actual}$

Sensor Quality / Noise Analysis — GPS behaves as expected

[INFO] GPS behaves as expected

What happened

GPS: mean err 3.94, σ 2.80, bias (0.12, 0.48).

What it probably means

GPS mean error and bias are consistent with the reported σ — no immediate noise / calibration concern.

Why it matters

This sensor is likely a reliable contributor to the fusion.

Confirmed evidence

- 13 GPS measurements compared to ground truth.
- Mean error = 3.94, max error = 9.28, RMSE = 4.60.
- Bias = ($\Delta x=0.12$, $\Delta y=0.48$).
- Mean reported σ = 2.80.

Sensor Quality / Noise Analysis — Camera behaves as expected

[INFO] Camera behaves as expected

What happened

Camera: mean err 0.39, σ 0.32, bias (0.04, 0.06).

What it probably means

Camera mean error and bias are consistent with the reported σ — no immediate noise / calibration concern.

Why it matters

This sensor is likely a reliable contributor to the fusion.

Confirmed evidence

- 40 Camera measurements compared to ground truth.
- Mean error = 0.39, max error = 0.88, RMSE = 0.44.
- Bias = ($\Delta x=0.04$, $\Delta y=0.06$).
- Mean reported σ = 0.32.

Sensor Quality / Noise Analysis — DAS behaves as expected

[INFO] DAS behaves as expected

What happened

DAS: mean err 0.45, σ 0.30, bias (-0.02, -0.04).

What it probably means

DAS mean error and bias are consistent with the reported σ — no immediate noise / calibration concern.

Why it matters

This sensor is likely a reliable contributor to the fusion.

Confirmed evidence

- 93 DAS measurements compared to ground truth.
- Mean error = 0.45, max error = 1.09, RMSE = 0.50.
- Bias = (Δx =-0.02, Δy =-0.04).
- Mean reported σ = 0.30.

Error Spike Investigation

Error Spike Investigation

Every detected spike or jump is paired with the local sensor context (what was available before / during / after) and a rule-based explanation that distinguishes confirmed facts from likely causes.

- No major error spikes were detected in any track.

Track Fragmentation Analysis

Track Fragmentation Analysis

1 oracle vehicle(s) were observed across 1 track ID(s). Each oracle vehicle maps to a single track — no fragmentation suspected.

Track Fragmentation Analysis — table

Vehicle	# tracks	Tracks	Filtered tracks
veh1	1	T000001 [0.30-12.00s, 235 rows]	—

Segment Transition Analysis

Segment Transition Analysis

Compares the segment transitions the report would emit against the transitions implied by oracle lane_id over time. Mismatches usually indicate a coverage / drift issue rather than a transition-logic bug.

Segment Transition Analysis — table

Track	Vehicle	Reported transitions	Oracle transitions	Match
T000001	veh1	seg1→seg2@7.40s	seg2	no

Segment Transition Analysis — Segment transition inconsistency

[WARN] Segment transition inconsistency

What happened

Track T000001: 1 reported vs 0 oracle transitions.

What it probably means

The report shows more transitions than the oracle indicates – the segment assignment is flickering between adjacent segments, probably because the estimated position is on a boundary.

Why it matters

Segment transitions are how the report tells the story of the vehicle's journey; missing or extra ones distort that story.

Confirmed evidence

- Reported transitions: 1 (seg1→seg2@7.40s).
- Oracle transitions inside this track's time window: 0 (seg2).
- Track time window: $t = 0.30\text{--}12.00$ s.

Possible explanations (hypotheses)

- Estimated position is oscillating across a segment boundary.
- Lane-to-segment mapping has overlapping centroids near the boundary.

Camera Geometry / FOV Analysis

Camera Geometry / FOV Analysis

Per-track camera measurement counts, time windows of camera presence, and a sanity check on the camera σ vs actual error. Conclusions about FOV are heuristic — the audit data does not carry the camera polygon itself.

Camera Geometry / FOV Analysis — table

Track	Cam meas	Cam window (s)	Pre-tr / post-tr	Mean σ	Mean err
T000001	40	11.20	29/11	0.32	0.39

Camera Geometry / FOV Analysis — Camera contributes mostly before the transition

[INFO] Camera contributes mostly before the transition

What happened

Camera meas pre/post first transition: 29/11.

What it probably means

The camera dominates the early segment and then largely disappears; the post-transition tracking depends on DAS.

Why it matters

Asymmetric camera coverage explains why error often rises right after a segment transition even when the filter is healthy.

Confirmed evidence

- Camera-window length: 11.20s with the bulk before the transition.

Possible explanations (hypotheses)

- FOV layout intentionally favors the early segment.

Recommended next checks

- Decide whether post-segment camera coverage is worth adding.

DAS Quality Analysis

DAS Quality Analysis

DAS measurement counts, SNR distribution, and the Kalman error in DAS-only windows. Where SNR data is available, correlations with tracking error are flagged as evidence of σ -from-SNR mis-calibration.

DAS Quality Analysis — table

Track	DAS meas	Mean SNR	Min SNR	Low-SNR rows	Mean err
T000001	93	103.33	32.52	0	0.31

Kalman Behavior Analysis

Kalman Behavior Analysis

For each track the Kalman is evaluated on: how often it received real updates (vs prediction), whether σ grows in gaps and shrinks after measurements, and whether the reported σ matches the actual error. Strong mismatches indicate a tuning issue — not a bug.

Kalman Behavior Analysis — table

Track	% pred-only	σ growth in gaps	σ shrink after meas	σ vs actual err
T000001	43.4	34/35	33/35	$\sigma \approx \text{err}$ (healthy)

Kalman Behavior Analysis — Kalman behaves correctly

[INFO] Kalman behaves correctly

What happened

σ tracks actual error, σ grows during gaps, shrinks after measurements.

What it probably means

All meaningful tracks show the textbook Kalman pattern: uncertainty rises while coasting and falls when sensor data returns, and the reported uncertainty matches the actual error well.

Why it matters

No tuning intervention needed for the filter itself.

Confirmed evidence

- 1/1 tracks classified healthy.

Recommendations & Next Investigations

Recommendations & Next Investigations

Concrete checks aggregated from the diagnostic sections. Items are ordered by severity (critical → warning → info) and de-duplicated. Each is linked to the interpretation that suggested it.

- (Segment transition inconsistency) Compare reported vs oracle segment IDs on the trajectory plot.
- (Segment transition inconsistency) Decide whether `segment_id` should be assigned from oracle position instead of estimated position for reporting purposes.
- (Camera contributes mostly before the transition) Decide whether post-segment camera coverage is worth adding.